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SIMATS ENGINEERING

ASSESSMENT TOOL 1

APPLICATION BASED QUESTIONS

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Sol) Given:

- Box A: 6 Red, 4 Blue, 2 Green (12 balls)
- Box B: 5 Red, 7 Blue, 3 Green (15 balls)
- Box C: 8 Red, 2 Blue, 5 Green (15 balls)
- Box B is twice as likely to be chosen.

Let:

- $P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$

Probability of drawing a blue ball:

- $P(\text{Blue} | A) = \frac{4}{12} = \frac{1}{3}$
- $P(\text{Blue} | B) = \frac{7}{15}$
- $P(\text{Blue} | C) = \frac{2}{15}$

Formula (Bayes' Theorem):

$$P(C | \text{Blue}) = \frac{P(C) \times P(\text{Blue} | C)}{P(A)P(\text{Blue} | A) + P(B)P(\text{Blue} | B) + P(C)P(\text{Blue} | C)}$$

Calculation:

Numerator

$$= \frac{1}{4} \times \frac{2}{15} = \frac{1}{30}$$

Denominator

$$\begin{aligned} &= \frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15} \\ &= \frac{1}{12} + \frac{7}{30} + \frac{1}{30} = \frac{7}{20} \\ P(C | \text{Blue}) &= \frac{\frac{1}{30}}{\frac{7}{20}} = \frac{2}{21} = 0.0952 \end{aligned}$$

Result:

$P(\text{Box C} \text{Blue}) = \frac{2}{21} \approx 0.0952 = 9.52\%$
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2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Sol) Given:

- Section A: 5 Red, 3 Blue, 2 Black (10 cars)
- Section B: 4 Red, 6 Blue, 5 Black (15 cars)
- Section C: 7 Red, 2 Blue, 6 Black (15 cars)

Selection probabilities:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Blue probabilities:

$$\begin{aligned} P(BL | A) &= \frac{3}{10} \\ P(BL | B) &= \frac{6}{15} = \frac{2}{5} \\ P(BL | C) &= \frac{2}{15} \end{aligned}$$

Formula

$$P(C | BL) = \frac{P(C)P(BL | C)}{P(A)P(BL | A) + P(B)P(BL | B) + P(C)P(BL | C)}$$

Calculation

Numerator

$$= \frac{1}{4} \times \frac{2}{15} = \frac{1}{30}$$

Denominator

$$\begin{aligned} &= \frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15} \\ &= \frac{3}{40} + \frac{1}{5} + \frac{1}{30} = \frac{37}{120} \\ P(C | BL) &= \frac{\frac{1}{30}}{\frac{37}{120}} = \frac{4}{37} = 0.1081 \end{aligned}$$

Result

$P(\text{Section C} \text{Blue}) = \frac{4}{37} \approx 0.1081 = 10.81\%$

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model

reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

Sol)

Given

Total samples per class = 1800

Test data = 20%

Test samples per class

$$1800 \times 20\% = 360$$

Positive class = 360

Negative class = 360

Given:

TP = 210

TN = 190

FP = 30

FN = ?

FN Calculation

$$FN = 360 - 210 = 150$$

Total test samples

$$360 + 360 = 720$$

Accuracy

$$= \frac{TP + TN}{Total} = \frac{210 + 190}{720} = \frac{400}{720} = 55.56\%$$

Precision

$$= \frac{TP}{TP + FP} = \frac{210}{210 + 30} = \frac{210}{240} = 87.5\%$$

Sensitivity (Recall)

$$= \frac{TP}{TP + FN} = \frac{210}{210 + 150} = \frac{210}{360} = 58.33\%$$

Specificity

$$= \frac{TN}{TN + FP} = \frac{190}{190 + 30} = \frac{190}{220} = 86.36\%$$

Result

- FN = 150
- Accuracy = 55.56%
- Precision = 87.50%
- Sensitivity = 58.33%
- Specificity = 86.36%

Effectiveness:

The model has high precision and specificity, meaning it predicts positive cases accurately and identifies healthy cases well. However, the recall is low (58.33%), so it misses many actual disease cases, which is undesirable for medical diagnosis.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0

(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1

(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3

(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9

(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.

ii) Suggest two effective techniques to reduce overfitting in this model.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Sol)

Given:

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

(i) Identify the epoch at which the model starts overfitting.

Observation:

- From Epoch 1 to Epoch 5, both training accuracy and validation accuracy increase.
- At Epoch 5, validation accuracy reaches its maximum (73%).
- From Epoch 6, training accuracy continues to increase (85%, 88%, 92%), but validation accuracy starts decreasing (72%, 70%, 68%).

This indicates that the model is learning the training data too well and its performance on unseen data is getting worse.

Answer:

The model starts overfitting at Epoch 6.
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(ii) Explain why validation accuracy decreases despite training accuracy increasing.

Explanation:

- The model begins to memorize the training data instead of learning general patterns.
- As a result, it performs very well on the training dataset.
- However, it cannot predict new (validation/test) data accurately.
- Therefore, training accuracy increases, while validation accuracy decreases.

Answer:

Validation accuracy decreases because the model overfits the training data, reducing its ability to generalize to unseen data.

(iii) Suggest two methods to improve validation performance.

1. Early Stopping

- Stop training when the validation accuracy stops improving (or validation loss starts increasing).
- This prevents the model from overfitting.

2. Dropout (Regularization)

- Randomly disables some neurons during training.
- This forces the model to learn more robust features and improves generalization.

Answer:

1. Early Stopping
2. Dropout (Regularization)

Final Answer

(i) The model starts overfitting at Epoch 6, because training accuracy keeps increasing while validation accuracy decreases.

(ii) Validation accuracy decreases because the model memorizes the training data instead of learning general patterns, leading to poor performance on unseen data.

(iii) Two methods to improve validation performance are:

- Early Stopping – Stops training before overfitting begins.
- Dropout (Regularization) – Prevents overfitting by randomly deactivating neurons during training

5. Examine the performance of a machine-learning model using training and validation

accuracy over multiple epochs:

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

(h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

iii) Suggest two methods to improve validation performance.

Sol)

Given

Epoch Training Loss Validation Loss

1	3.0	2.7
2	2.6	2.3
3	2.3	2.1
4	2.1	2.0
5	1.9	2.2
6	1.7	2.5
7	1.6	2.8
8	1.4	3.1

(i) At which epoch does the model achieve optimal generalization?

The model achieves optimal generalization at Epoch 4 because the validation loss is minimum (2.0) at this epoch. After Epoch 4, the validation loss starts increasing.

Optimal Generalization = Epoch 4

(ii) Identify signs of overfitting from the data.

The signs of overfitting are:

- Training loss decreases continuously (from 2.1 to 1.4).
- Validation loss increases after Epoch 4 (from 2.0 to 3.1).

This shows that the model is fitting the training data too closely and its performance on unseen data is getting worse.

(iii) Recommend two regularization techniques and explain their impact.

1. Dropout

- Randomly deactivates some neurons during training.
- Prevents overfitting and improves the model's generalization.

2. L2 Regularization (Weight Decay)

- Adds a penalty for large weights in the loss function.
- Reduces model complexity and helps prevent overfitting.

6. A machine learning model shows the following loss values during training:

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7

(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3

(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1

(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0

(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2

(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5

(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8

(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1

i) At which epoch does the model achieve optimal generalization?

ii) Identify signs of overfitting from the data.

iii) Recommend two regularization techniques and explain their impact.

Sol)

Given

Epoch	Training Loss	Validation Loss
1	3.0	2.7
2	2.6	2.3
3	2.3	2.1
4	2.1	2.0
5	1.9	2.2
6	1.7	2.5
7	1.6	2.8
8	1.4	3.1

(i) At which epoch does the model achieve optimal generalization?

Observation:

- Validation loss decreases from 2.7 to 2.0 (Epoch 1 → Epoch 4).
- The minimum validation loss is 2.0 at Epoch 4.
- After Epoch 4, the validation loss increases.

Answer:

The model achieves optimal generalization at Epoch 4.

(ii) Identify signs of overfitting from the data.

Observation:

- Training loss keeps decreasing:
 - 2.1 → 1.9 → 1.7 → 1.6 → 1.4
- Validation loss starts increasing:

- 2.0 → 2.2 → 2.5 → 2.8 → 3.1

Answer:

The signs of overfitting are:

1. Training loss continuously decreases.
2. Validation loss starts increasing after Epoch 4.

This indicates that the model is learning the training data too closely and is performing worse on unseen data.

(iii) Recommend two regularization techniques and explain their impact.

1. Dropout

- Randomly deactivates some neurons during training.
- Prevents overfitting and improves the model's ability to generalize.

2. L2 Regularization (Weight Decay)

- Adds a penalty to large weights in the model.
- Reduces model complexity and prevents overfitting.

Final Answers

(i) The model achieves optimal generalization at Epoch 4, because the validation loss is the lowest (2.0).

(ii) The signs of overfitting are:

- Training loss decreases continuously.
- Validation loss increases after Epoch 4.

(iii) Two regularization techniques are:

1. Dropout – Prevents overfitting by randomly disabling neurons during training.
2. L2 Regularization – Penalizes large weights, reducing overfitting and improving generalization.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires

automatically. Let the number of tires are 250 (125 front and 125 back tires).

a) Front identified as Front: 119

b) Front identifies as Back: 06

c) Back identified as Back: 120

d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Sol)

Given:

- Total tires = **250**
- Front tires = **125**
- Back tires = **125**

Confusion Matrix:

- **True Positive (TP)** = Front identified as Front = **119**
- **False Negative (FN)** = Front identified as Back = **6**
- **True Negative (TN)** = Back identified as Back = **120**
- **False Positive (FP)** = Back identified as Front = **5**

1. Accuracy

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{119 + 120}{119 + 120 + 5 + 6} \times 100 \\ &= \frac{239}{250} \times 100 \\ &= 95.6\% \end{aligned}$$

Answer:

$\text{Accuracy} = 95.6\%$

2. Precision

Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{119}{119 + 5} \times 100 \\ &= \frac{119}{124} \times 100 \\ &= 95.97\% \end{aligned}$$

Answer:

$$\boxed{\text{Precision} = 95.97\%}$$

3. Sensitivity (Recall)

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{119}{119 + 6} \times 100 \\ &= \frac{119}{125} \times 100 \\ &= 95.2\% \end{aligned}$$

Answer:

$$\boxed{\text{Sensitivity} = 95.2\%}$$

4. Specificity

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Calculation:

$$\begin{aligned}
&= \frac{120}{120 + 5} \times 100 \\
&= \frac{120}{125} \times 100 \\
&= 96.0\%
\end{aligned}$$

Answer:

$$\boxed{\text{Specificity} = 96.0\%}$$

5. F1-Score

Formula:

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN} \times 100$$

Calculation:

$$\begin{aligned}
&= \frac{2 \times 119}{2 \times 119 + 5 + 6} \times 100 \\
&= \frac{238}{249} \times 100 \\
&= 95.58\%
\end{aligned}$$

Answer:

$$\boxed{\text{F1-Score} = 95.58\%}$$

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138

b) Diseased identified as Healthy: 12

c) Healthy identified as Healthy: 135

d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Sol)

Given

- Total cases = 300
- Diseased = 150
- Healthy = 150

Confusion Matrix:

- True Positive (TP) = Diseased identified as Diseased = 138
- False Negative (FN) = Diseased identified as Healthy = 12
- True Negative (TN) = Healthy identified as Healthy = 135
- False Positive (FP) = Healthy identified as Diseased = 15

1. Accuracy

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{138 + 135}{138 + 135 + 15 + 12} \times 100 \\ &= \frac{273}{300} \times 100 \\ &= 91.0\% \end{aligned}$$

Answer:

Accuracy = 91.0%

2. Precision

Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{138}{138 + 15} \times 100 \\
 &= \frac{138}{153} \times 100 \\
 &= 90.20\%
 \end{aligned}$$

Answer:

Precision = 90.20%

3. Sensitivity (Recall)

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{138}{138 + 12} \times 100 \\
 &= \frac{138}{150} \times 100 \\
 &= 92.0\%
 \end{aligned}$$

Answer:

Sensitivity = 92.0%

4. Specificity

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{135}{135 + 15} \times 100 \\
 &= \frac{135}{150} \times 100 \\
 &= 90.0\%
 \end{aligned}$$

Answer:

$$\boxed{\text{Specificity} = 90.0\%}$$

5. F1-Score

Formula:

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{2 \times 138}{2 \times 138 + 15 + 12} \times 100 \\ &= \frac{276}{303} \times 100 \\ &= 91.09\% \end{aligned}$$

Answer:

$$\boxed{\text{F1-Score} = 91.09\%}$$

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Sol)

Given

- Total emails = 400
- Spam emails = 200
- Non-Spam emails = 200

Confusion Matrix:

- True Positive (TP) = Spam identified as Spam = 180

- False Negative (FN) = Spam identified as Not Spam = 20
- True Negative (TN) = Not Spam identified as Not Spam = 170
- False Positive (FP) = Not Spam identified as Spam = 30

1. Accuracy

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{180 + 170}{180 + 170 + 30 + 20} \times 100 \\ &= \frac{350}{400} \times 100 \\ &= 87.5\% \end{aligned}$$

Answer:

Accuracy = 87.5%

2. Precision

Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{180}{180 + 30} \times 100 \\ &= \frac{180}{210} \times 100 \\ &= 85.71\% \end{aligned}$$

Answer:

Precision = 85.71%

3. Sensitivity (Recall)

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{180}{180 + 20} \times 100 \\ &= \frac{180}{200} \times 100 \\ &= 90.0\% \end{aligned}$$

Answer:

Sensitivity = 90.0%

4. Specificity

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{170}{170 + 30} \times 100 \\ &= \frac{170}{200} \times 100 \\ &= 85.0\% \end{aligned}$$

Answer:

Specificity = 85.0%

5. F1-Score

Formula:

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{2 \times 180}{2 \times 180 + 30 + 20} \times 100 \\
 &= \frac{360}{410} \times 100 \\
 &= 87.80\%
 \end{aligned}$$

Answer:

$$\boxed{\text{F1-Score} = 87.80\%}$$

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 230
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Sol)

Given

- Total products = 500
- Defective products = 250
- Non-Defective products = 250

Confusion Matrix:

- True Positive (TP) = Defective identified as Defective = 230
- False Negative (FN) = Defective identified as Non-Defective = 20
- True Negative (TN) = Non-Defective identified as Non-Defective = 225
- False Positive (FP) = Non-Defective identified as Defective = 25

1. Accuracy

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{230 + 225}{230 + 225 + 25 + 20} \times 100 \\ &= \frac{455}{500} \times 100 \\ &= 91.0\% \end{aligned}$$

Answer:

$$\boxed{\text{Accuracy} = 91.0\%}$$

2. Precision

Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{230}{230 + 25} \times 100 \\ &= \frac{230}{255} \times 100 \\ &= 90.20\% \end{aligned}$$

Answer:

$$\boxed{\text{Precision} = 90.20\%}$$

3. Sensitivity (Recall)

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{230}{230 + 20} \times 100 \\
 &= \frac{230}{250} \times 100 \\
 &= 92.0\%
 \end{aligned}$$

Answer:

$$\boxed{\text{Sensitivity} = 92.0\%}$$

4. Specificity

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{225}{225 + 25} \times 100 \\
 &= \frac{225}{250} \times 100 \\
 &= 90.0\%
 \end{aligned}$$

Answer:

$$\boxed{\text{Specificity} = 90.0\%}$$

5. F1-Score

Formula:

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN} \times 100$$

Calculation:

$$\begin{aligned}
 &= \frac{2 \times 230}{2 \times 230 + 25 + 20} \times 100 \\
 &= \frac{460}{505} \times 100 \\
 &= 91.09\%
 \end{aligned}$$

Answer:

$$\boxed{\text{F1-Score} = 91.09\%}$$

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

Sol)

Given

- **Shop X:**
 - Pure oil = 20
 - Adulterated oil = 30
 - Total bottles = 50
- **Shop Y:**
 - Pure oil = 40
 - Adulterated oil = 20
 - Total bottles = 60
- **Shop Y is twice as likely to be chosen as Shop X.**

Therefore,

$$P(X) = \frac{1}{3}, P(Y) = \frac{2}{3}$$

Probability of selecting an adulterated bottle:

$$P(A | X) = \frac{30}{50} = \frac{3}{5}$$
$$P(A | Y) = \frac{20}{60} = \frac{1}{3}$$

Formula (Bayes' Theorem)

$$P(Y | A) = \frac{P(Y) \times P(A | Y)}{P(X) \times P(A | X) + P(Y) \times P(A | Y)}$$

Calculation

Step 1: Numerator

$$P(Y) \times P(A | Y) = \frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$$

Step 2: Denominator

$$\begin{aligned} &= \frac{1}{3} \times \frac{3}{5} + \frac{2}{3} \times \frac{1}{3} \\ &= \frac{1}{5} + \frac{2}{9} \end{aligned}$$

Taking LCM = 45,

$$= \frac{9}{45} + \frac{10}{45} = \frac{19}{45}$$

Step 3: Apply Bayes' Theorem

$$\begin{aligned} P(Y | A) &= \frac{\frac{2}{9}}{\frac{19}{45}} \\ &= \frac{2}{9} \times \frac{45}{19} \\ &= \frac{10}{19} \\ &= \frac{10}{19} = 0.5263 \end{aligned}$$

Final Answer

$$P(\text{Shop Y} | \text{Adulterated}) = \frac{10}{19} \approx 0.5263$$

Therefore, the probability that the adulterated bottle came from Shop Y is:

$$0.5263 \text{ or } 52.63\%$$

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

Sol)

Given

- **Section A:**
 - Defective items = 15
 - Non-defective items = 35
 - Total items = 50
- **Section B:**
 - Defective items = 25
 - Non-defective items = 25
 - Total items = 50
- **Section A is three times as likely to be chosen as Section B.**

Therefore,

$$P(A) = \frac{3}{4}, P(B) = \frac{1}{4}$$

Probability of selecting a defective item:

$$P(D | A) = \frac{15}{50} = \frac{3}{10}$$
$$P(D | B) = \frac{25}{50} = \frac{1}{2}$$

Formula (Bayes' Theorem)

$$P(B | D) = \frac{P(B) \times P(D | B)}{P(A) \times P(D | A) + P(B) \times P(D | B)}$$

Calculation

Step 1: Numerator

$$P(B) \times P(D | B) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

Step 2: Denominator

$$\begin{aligned} &= \frac{3}{4} \times \frac{3}{10} + \frac{1}{4} \times \frac{1}{2} \\ &= \frac{9}{40} + \frac{5}{40} = \frac{14}{40} = \frac{7}{20} \end{aligned}$$

Step 3: Apply Bayes' Theorem

$$\begin{aligned} P(B | D) &= \frac{\frac{1}{8}}{\frac{7}{20}} \\ &= \frac{1}{8} \times \frac{20}{7} \\ &= \frac{5}{14} \\ &= \frac{5}{14} = 0.3571 \end{aligned}$$

Final Answer

$$P(\text{Section B} | \text{Defective}) = \frac{5}{14} \approx 0.3571$$

Therefore, the probability that the defective item came from Section B is:

$$\boxed{0.3571 \text{ or } 35.71\%}$$

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing.

Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Sol)

Given

- Total samples per class = 1000
- Test data = 20%

Therefore,

- Positive class (Actual Positive) = $1000 \times 20\% = 200$
- Negative class (Actual Negative) = $1000 \times 20\% = 200$

Given:

- True Positive (TP) = 120
- True Negative (TN) = 150

Step 1: Find FN and FP

False Negative (FN)

$$FN = 200 - TP = 200 - 120 = 80$$

False Positive (FP)

$$FP = 200 - TN = 200 - 150 = 50$$

1. Accuracy

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{120 + 150}{120 + 150 + 50 + 80} \times 100 \\ &= \frac{270}{400} \times 100 \\ &= 67.5\% \end{aligned}$$

Answer:

$$\boxed{\text{Accuracy} = 67.5\%}$$

2. Precision

Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{120}{120 + 50} \times 100 \\ &= \frac{120}{170} \times 100 \\ &= 70.59\% \end{aligned}$$

Answer:

$$\boxed{\text{Precision} = 70.59\%}$$

3. Sensitivity (Recall)

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{120}{120 + 80} \times 100 \\ &= \frac{120}{200} \times 100 \\ &= 60.0\% \end{aligned}$$

Answer:

$$\boxed{\text{Sensitivity} = 60.0\%}$$

4. Specificity

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Calculation:

$$\begin{aligned} &= \frac{150}{150 + 50} \times 100 \\ &= \frac{150}{200} \times 100 \\ &= 75.0\% \end{aligned}$$

Answer:

$$\boxed{\text{Specificity} = 75.0\%}$$

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.

Calculate: Accuracy Precision Sensitivity Specificity.

Sol)

Given:

Epoch Training Loss Validation Loss

1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

(i) Identify the epoch at which the model begins to overfit.

Observation:

- Training loss decreases continuously: $2.8 \rightarrow 1.3$
- Validation loss decreases only up to Epoch 3 (2.0).

- From Epoch 4, validation loss starts increasing (2.1, 2.3, 2.6, 2.9, 3.2).

Answer:

The model begins to overfit at Epoch 4.

(ii) Suggest two effective techniques to reduce overfitting:

1. Early Stopping

- Stops training when validation loss starts increasing.
- Prevents the model from overfitting.

2. Dropout

- Randomly deactivates some neurons during training.
- Reduces overfitting and improves generalization.

(iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result, the model performs well on the training data but poorly on unseen (validation/test) data, reducing its generalization ability.

Final Answers

(i) The model begins to overfit at Epoch 4, because validation loss starts increasing while training loss continues decreasing.

(ii) Two techniques to reduce overfitting are:

1. Early Stopping
2. Dropout

(iii) Overfitting reduces the model's ability to generalize to unseen data, leading to poor performance on validation and test datasets.

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6

ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2

- iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
- iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
- v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
- vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
- vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
- viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

- a. At which epoch does the model achieve the best generalization performance?
- b. At which point does overfitting begin? Justify your answer.
- c. Suggest two techniques to reduce overfitting and briefly explain how they help.

Sol)

Given

Epoch	Training Loss	Validation Loss
1	2.8	2.6
2	2.4	2.2
3	2.1	1.9
4	1.9	1.8
5	1.7	1.9
6	1.5	2.1
7	1.3	2.4
8	1.2	2.7

(a) At which epoch does the model achieve the best generalization performance?

Observation:

- Validation loss decreases from 2.6 to 1.8.
- The lowest validation loss is 1.8 at Epoch 4.

- After Epoch 4, validation loss starts increasing.

Answer:

The model achieves the best generalization performance at Epoch 4.

(b) At which point does overfitting begin? Justify your answer.

Observation:

- Training loss continues to decrease after Epoch 4 (1.9 → 1.2).
- Validation loss starts increasing after Epoch 4 (1.8 → 1.9 → 2.1 → 2.4 → 2.7).

Answer:

Overfitting begins at Epoch 5.

Justification:

From Epoch 5, the training loss continues to decrease while the validation loss increases. This indicates that the model is memorizing the training data and performing worse on unseen data.

(c) Suggest two techniques to reduce overfitting and briefly explain how they help.

1. Early Stopping

- Stops training when validation loss starts increasing.
- Prevents the model from overfitting and improves generalization.

2. Dropout

- Randomly deactivates some neurons during training.
- Prevents the model from depending too much on specific neurons and reduces overfitting.

Final Answers

(a) The model achieves the best generalization performance at Epoch 4, because the validation loss is minimum (1.8).

(b) Overfitting begins at Epoch 5, because training loss keeps decreasing while validation loss starts increasing.

(c) Two techniques to reduce overfitting are:

1. Early Stopping – Stops training before overfitting begins.

2. Dropout – Randomly disables neurons during training to improve generalization.